

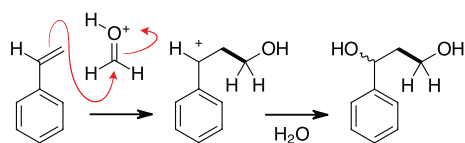
Personal Exploration Plan

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Introducing the topic and the plan for my personal literature exploration.

The Prins Reaction

I had become aware of the Prins reaction when reading a recent paper reporting a total synthesis of a natural product.¹ The Prins reaction was first observed by Henrick Prins in 1919. It is a carbon-carbon bond-forming reaction between an activated carbonyl group and an alkene.² The mechanism is a specific acid-catalyzed electrophilic addition. I searched the literature using SciFinder and Google Scholar and found a series of papers published in the early 1970s by Fernand Coussement. He investigated the mechanism and reported the results of Hammett plot experiments. In the abstract of the 18th paper in the series, Coussement reports evidence for a nonclassical carbocationic intermediate.³ There seems to be the possibility of data to plot and interesting ideas to explore. Unfortunately, all his work is in French and in a defunct journal that I cannot easily access.



Another earlier series of papers by Lloyd Dolby was more accessible. A Hammett plot was part of these experiments.⁴ There are also experiments using isotopic labelling⁵ and evidence of the nonclassical cation.⁶

My Plan

I plan to describe the three experiments reported in the abovementioned Dolby papers. I will extract the data for the Hammett plot and perform a line fit to see if I get the exact numbers. I will replace the σ values used by the authors with the modern data set and compare the results. I will also discuss the idea of the nonclassical carbocation intermediate and how the experimental evidence supports that hypothesis.

The Prins reaction is not presented in the textbook, but electrophilic addition reactions are well covered. The nonclassical cation is also discussed in detail – it's a famous story in physical organic chemistry. I look forward to learning more about it.

I will get access to the Coussement papers and translate enough to interpret his Hammett plots and compare them to Dolby's results. I will soon be taking a walk to speak with a librarian.⁷

This document was produced using the L^AT_EX typesetting language with the Tufte-handout document class. Chemical diagrams were created in ChemDoodle and modified using Affinity Designer.

¹ "A Biomimetic Approach for Construction of an Advanced Intermediate en Route to Melicolones A and B." T. Wang, Z. Jia, C. Zhu, L. Zhang, H. Ding, *J. Org. Chem.*, 2025, 90, 7531–7539. <https://doi.org/10.1021/acs.joc.5c00734>.

² For a general overview of the Prins reaction and recent examples see "Recent Advances in the Prins Reaction." E. Reyes, L. Prieto, U. Uria, L. Carrillo, J.L. Vicario, *ACS Omega*, 2022, 7, 31621–31627. <https://doi.org/10.1021/acsomega.2c04765>.

³ "Prins reaction. XVIII. Influence of the acidity and of the meta or para substituents on the reactivity of β -methylstyrenes." C. Bocard, M. Davidson, M. Hellin, F. Coussement, *Bull. Soc. Chim. France*, 1971, 163–169.

Figure 1: The Prins reaction.

⁴ "Mechanism of the Prins reaction. VII. Kinetic studies of the Prins reaction of styrenes." L.J. Dolby, C.L. Wilkins, R.M. Rodia, *J. Org. Chem.*, 1968, 33, 4155–4158. <https://doi.org/10.1021/jo01275a031>.

⁵ "The Mechanism of the Prins Reaction. IV. Evidence against Acetoxonium Ion Intermediates." L.J. Dolby, M.J. Schwarz, *J. Org. Chem.*, 1965, 30, 3581–3586. <https://doi.org/10.1021/jo01022a001>

⁶ "The Mechanism of the Prins Reaction. III. The Acetolysis of Arenesulfonates of anti-3-Oxabicyclo[3.3.1]nonan-9-ol and trans-6-Hydroxy-cis-3-oxabicyclo[4.3.0]nonane." L.J. Dolby, M.J. Schwarz, *J. Org. Chem.*, 1963, 28, 1456–1458. <https://doi.org/10.1021/jo01041a003>

⁷ UPDATE: The librarian informed me that the *Bulletin de la Société Chimique de France* is available in the UPEI bound periodical stacks (for 1967 to 1984). They shamed me for assuming that if it's not online, then it is nowhere.